# MTE 204 – Numerical Methods
# Tutorial #4 Solutions

## Question 1:

An investigator has reported the data tabulated below for an experiment to determine the growth rate of bacteria $k$ (per day) as a function of oxygen concentration $c$ (mg/L). It is known that such data can be modeled by the following equation:

$$k = \frac{k_{max} c^2}{c_s + c^2}$$

Where $c_s$ and $k_{max}$ are constants.

| $c$ | 0.5 | 0.8 | 1.5 | 2.5 | 4 |
|---|---|---|---|---|---|
| $k$ | 1.1 | 2.4 | 5.3 | 7.6 | 8.9 |

(a) Use a transformation to linearize this equation
(b) Use linear regression to estimate $c_s$ and $k_{max}$
(c) Predict the growth rate at $c$ = 2 mg/L
(d) Calculate the coefficient of determination ($r^2$) and the sum of the squares of the residuals ($S_r$)

## Solution:

## Part a:

The equation $k = \frac{k_{max} c^2}{c_s + c^2}$ can be linearized by inverting it to give:

$$\frac{1}{k} = \frac{c_s + c^2}{k_{max} c^2} = \frac{1}{k_{max}} + \frac{c_s}{k_{max}} \frac{1}{c^2}$$

Thus, a plot of $\frac{1}{k}$ versus $\frac{1}{c^2}$ will be linear, with a slope of $\frac{c_s}{k_{max}}$ and an intercept of $\frac{1}{k_{max}}$.

The linearized equation can be expressed as

$$y = a_0 + a_1 x$$

Where $y = \frac{1}{k}$, $x = \frac{1}{c^2}$, $a_0 = \frac{1}{k_{max}}$, and $a_1 = \frac{c_s}{k_{max}}$

## Part b:

We can use the normal equations and solve them simultaneously to determine the values of $a_0$ and $a_1$, which will allow us to estimate $c_s$ and $k_{max}$:

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - \left(\sum x_i\right)^2}$$

$$a_0 = \bar{y} - a_1\bar{x}$$

Next, we undertake the computations required to find the values of $a_0$ and $a_1$:

| i | $c_i$ | $k_i$ | $x_i = \frac{1}{c_i^2}$ | $y_i = \frac{1}{k_i}$ | $x_i y_i$ | $x_i^2$ | $y_i^2$ |
|---|---|---|---|---|---|---|---|
| 1 | 0.5 | 1.1 | 4.000 | 0.9091 | 3.6364 | 16.0000 | 0.8264 |
| 2 | 0.8 | 2.4 | 1.5625 | 0.4167 | 0.6510 | 2.4414 | 0.1736 |
| 3 | 1.5 | 5.3 | 0.4444 | 0.1887 | 0.0839 | 0.1975 | 0.0356 |
| 4 | 2.5 | 7.6 | 0.1600 | 0.1316 | 0.0211 | 0.0256 | 0.0173 |
| 5 | 4 | 8.9 | 0.0625 | 0.1124 | 0.0070 | 0.0039 | 0.0126 |
| $\sum$ | 9.3 | 25.3 | 6.2294 | 1.7584 | 4.3993 | 18.6684 | 1.0656 |

$\sum x_i = 6.2294$ $\quad$ $\sum y_i = 1.7584$ $\quad$ $\sum x_i y_i = 4.3993$

${\sum x_i}^2 = 18.6684$ $\quad$ $\bar{x} = 6.2294/5 = 1.2459$ $\quad$ $\bar{y} = 1.7584/5 = 0.3517$

${\sum y_i}^2 = 1.0656$ $\quad$ $n = 5$

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - \left(\sum x_i\right)^2} = \frac{5(4.3993) - (6.2294 * 1.7584)}{5(18.6684) - (6.2294)^2}$$

$$a_1 = \frac{11.0427}{54.5365} = 0.2024$$

$$a_0 = \bar{y} - a_1\bar{x} = 0.3517 - (0.2024*1.2459) = 0.0995$$

Thus, the mathematical expression for the straight line is

$$y = 0.0995 + 0.2024x$$

We can estimate the values of $k_{max}$ and $c_s$ $using$:

$$a_0 = \frac{1}{k_{max}} \rightarrow k_{max} = \frac{1}{0.0995} = 10.0502$$

$$a_1 = \frac{c_s}{k_{max}} \rightarrow c_s = (0.2024 * 10.0502) = 2.0342$$

## Part c:

To predict the growth rate at $c$ = 2 mg/L, we need to find the value of $x$ corresponding to this value of $c$:

$$x = \frac{1}{c^2} = \frac{1}{2^2} = 0.25$$

$$y = 0.0995 + (0.2024 * 0.25) = 0.1501$$

Then, we can find $k$ using $y = \frac{1}{k}$

$$k = \frac{1}{0.1501} = 6.6622$$

*Note that you can also evaluate for $c$ = 2 mg/L directly in

$$k = \frac{k_{max} c^2}{c_s + c^2} = \frac{10.0502 * 2^2}{2.0342 + 2^2} = 6.6622$$

## Part d:

First, we calculate the correlation coefficient:

$$r = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x)^2} \sqrt{n \sum y_i^2 - (\sum y)^2}}$$

$$r = \frac{(5 * 4.3993) - (6.2294 * 1.7584)}{\sqrt{(5 * 18.6684) - 6.2294^2} \sqrt{(5 * 1.0656) - 1.7584^2}}$$

$$r = \frac{11.0427}{7.3848 * 1.4953} = 0.9999$$

Then, the coefficient of determination is

$$r^2 = 0.9999^2 = 0.9998$$

$r^2$ indicates that 99.98% of the original uncertainty has been explained by the linear model.

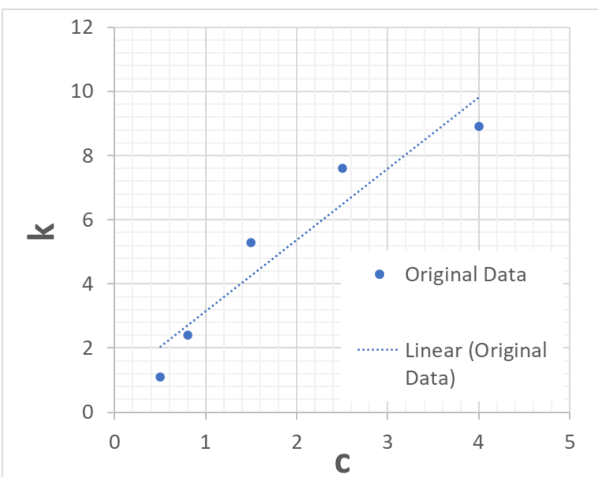
To find the standard error of the estimate, calculate the sum of the squares of the residuals:

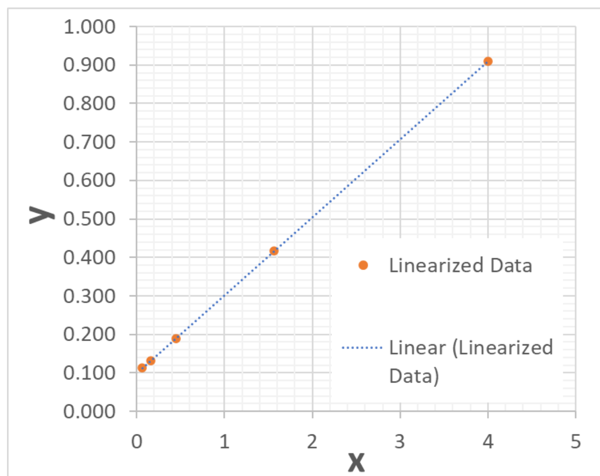
| i | $x_i$ | $y_i$ | Squares of the residuals $(y_i - a_0 - a_1 x_i)^2$ |
|---|---|---|---|
| 1 | 4.0000 | 0.9091 | 8.26446E-11 |
| 2 | 1.5625 | 0.4167 | 8.40278E-07 |
| 3 | 0.4444 | 0.1887 | 6.02658E-07 |
| 4 | 0.1600 | 0.1316 | 9.30571E-08 |
| 5 | 0.0625 | 0.1124 | 4.39114E-08 |
| $\sum$ | 6.2294 | 1.7584 | 1.58E-6 |

The sum of the squares of the residuals

$$S_r = 1.58 * 10^{-6}$$

*Note: The linear regression fit can be visualized before and after linearization using excel or MATLAB:



## Question 2:

The following data points were measured for an engineering experiment:

| $x$ | 1.6 | 2 | 2.5 | 3.2 |
|---|---|---|---|---|
| $f(x)$ | 2 | 8 | 14 | 15 |

(a) Develop quadratic splines for the data points and predict $f(2.2)$
(b) Compare your prediction to the prediction of a first-order spline

## Solution:

## Part a:

For the present problem, we have four data points and three intervals as shown in the table below.

| i | $x_i$ | $f_i$ |
|---|---|---|
| 1 | 1.6 | 2 |
| 2 | 2 | 8 |
| 3 | 2.5 | 14 |
| 4 | 3.2 | 15 |

The number of conditions required is (2 (n - 1) – 1), where n is the number of data points. This means that the number of conditions required is 5.

The polynomial for each interval can be expressed as

$$s_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2$$

Let $h$ define the width of the interval -> $h_i = x_{i+1} - x_i$

First, the function must pass through all points ($x = x_i \rightarrow h_i = 0$) to satisfy continuity condition:

$$f_i = a_i + b_i h_i + c_i {h_i}^2 \text{ simplifies to } f_i = a_i \qquad \text{Eqn. (18.32)}$$

Second, the function values of the adjacent polynomials must be equal

$$f_i + b_i h_i + c_i {h_i}^2 = f_{i+1} \qquad \text{Eqn. (18.34)}$$

Third, the first derivatives at the interior nodes must be equal:

$$b_i + 2c_i h_i = b_{i+1} \qquad \text{Eqn. (18.35)}$$

Forth, assume that the second derivative is zero at the first point

$$2c_1 = 0 \rightarrow \ c_1 = 0$$

➔ Therefore, the conditions required are:

$$f_i + b_i h_i + c_i {h_i}^2 = f_{i+1}$$ (<u>written for each interval→ three conditions)</u>

$$f_1 + b_1 h_1 = f_2 \qquad (1)$$

$$f_2 + b_2 h_2 + c_2 {h_2}^2 = f_3 \qquad (2)$$

$$f_3 + b_3 h_3 + c_3 {h_3}^2 = f_4 \qquad (3)$$

$$b_i + 2c_i h_i = b_{i+1}$$ (<u>written for the number of intervals minus one, two conditions)</u>

$$b_1 = b_2 \qquad (4)$$

$$b_2 + 2c_2 h_2 = b_3 \quad (5)$$

Rearrange equations (1) to (5):

$$h_1 b_1 = f_2 - f_1$$
$$h_2 b_2 + {h_2}^2 c_2 = f_3 - f_2$$
$$h_3 b_3 + {h_3}^2 c_3 = f_4 - f_3$$
$$b_1 - b_2 = 0$$
$$b_2 \ + 2h_2 c_2 - b_3 = 0$$

The interval widths are calculated using $h_i = \ x_{i+1} - x_i$:

| i | $\boldsymbol{x_i}$ | $\boldsymbol{f_i}$ | $\boldsymbol{h_i}$ |
|---|---|---|---|
| 1 | 1.6 | 2 | 0.4 |
| 2 | 2 | 8 | 0.5 |
| 3 | 2.5 | 14 | 0.7 |
| 4 | 3.2 | 15 | - |

After substituting these values into the conditions:

$$0.4\ b_1 = 6$$
$$0.5\ b_2 + 0.25\ c_2 = 6$$
$$0.7\ b_3 + 0.49\ c_3 = 1$$
$$b_1 - b_2 = 0$$
$$b_2 \ + c_2 - b_3 = 0$$

The system has five equations with five unknowns ( $b_1, b_2, c_2, b_3, c_3$ ), which can be expressed in matrix form. The coefficient matrix [A] can be expressed as

$$\begin{bmatrix} 0.4 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0.7 & 0.49 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \end{bmatrix}$$

The constants vector {B} is

$$\begin{Bmatrix} 6 \\ 6 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

and the unknowns' vector {x} is

$$\begin{Bmatrix} b_1 \\ b_2 \\ c_2 \\ b_3 \\ c_3 \end{Bmatrix}$$

The system can be solved using one of the techniques we learned for solving a system of linear equations (e.g., Gauss-Seidel or Gauss elimination methods):

$$\begin{bmatrix} 0.4 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0.7 & 0.49 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \\ c_2 \\ b_3 \\ c_3 \end{Bmatrix} = \begin{Bmatrix} 6 \\ 6 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

The roots of the system are:

$$\begin{Bmatrix} b_1 \\ b_2 \\ c_2 \\ b_3 \\ c_3 \end{Bmatrix} = \begin{Bmatrix} 15 \\ 15 \\ -6 \\ 9 \\ -10.8163 \end{Bmatrix}$$

Now, we can substitute the values for the a's, b's and c's into the quadratic spline function for each interval as follows

$$s_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2$$

Interval (1):

$$s_1(x) = a_1 + b_1(x - x_1) \qquad \text{(Recall } c_1 = 0\text{)}$$

$$s_1(x) = 2 + 15(x - 1.6)$$

Interval (2):

$$s_2(x) = a_2 + b_2(x - x_2) + c_2(x - x_2)^2$$

$$s_2(x) = 8 + 15(x - 2) - 6(x - 2)^2$$

Interval (3):

$$s_3(x) = a_3 + b_3(x - x_3) + c_3(x - x_3)^2$$

$$s_3(x) = 14 + 9(x - 2.5) - 10.8163(x - 2.5)^2$$

$x$ = 2.2 lies in the second interval between $x$ = 2 and $x$ = 2.5. Therefore, we use the spline function $s_2(x)$ to predict the value at x = 2.2:

$$s_2(2.2) = 8 + 15(2.2 - 2) - 6(2.2 - 2)^2 = 8 + 3 - 0.24 = 10.76$$

Note: Quadratic splines have their limitations, such as a) the first interval is represented by a straight line because of the assumption of $c_1 = 0$ and (2) the last interval may swing too high.

## Part b:

To evaluate the function at $x$ = 2.2 using a first order spline, we can use following equation to estimate the intermediate value between two points ($x_0$ = 2 and $x_1$ = 2.5):

$$f_2(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

$$f_2(2.2) = 8 + \frac{14 - 8}{2.5 - 2}(2.2 - 2)$$

$$f_2(2.2) = 10.4$$

It is clear that quadratic splines smooth out the abrupt slope change at the data points.